

Riemannian Gauss–Newton Method for Open-System Projected Variational Quantum Dynamics

1st Shiwen An

Dept. of Information and Communication Engineering
School of Engineering, Institute of Science Tokyo
Tokyo, Japan
an.s.0e57@m.isct.ac.jp

2nd Konstantinos Slavakis

Dept. of Information and Communication Engineering
School of Engineering, Institute of Science Tokyo
Tokyo, Japan
slavakis@ict.eng.isct.ac.jp

Abstract—Projected variational quantum dynamics (p-VQD) refits a fixed-depth ansatz to each Trotter step through a pure-state fidelity. That objective is undefined for the mixed state of an open system, and its obvious repair, the reduced trace overlap, is provably capped below unity for every optimizer. We purify the thermal bath with a thermofield ancilla. The enlarged system–bath–ancilla state stays pure, the fidelity applies exactly on the joint state, and tracing out bath and ancilla returns the physical reduced dynamics. Each projection is then a pure-state least-squares fit that a Riemannian Gauss–Newton inner loop with Trust-Region damping drives to the ansatz ceiling in a few iterations, where first-order p-VQD stalls orders of magnitude short. On finite-temperature spin–boson and central-spin benchmarks, validated against an independent TDVP baseline, the ansatz sets the attainable fidelity while the optimizer sets the convergence rate. Where the ansatz is under-resourced, the second-order step attains the best tested fidelities.

Index Terms—open quantum systems, variational quantum simulation, projected variational quantum dynamics, Gauss–Newton method, thermofield purification

I. INTRODUCTION

Projected variational quantum dynamics (p-VQD) [1] simulates real-time evolution at fixed circuit depth. It evolves the state one short Trotter step and refits the ansatz by maximizing a pure-state fidelity. In an *open* system the target is the mixed reduced state $\rho_S = \text{Tr}_E |\psi\rangle\langle\psi|$, and two obstacles follow. First, a mixed state has no pure-state fidelity, so the objective is undefined. Second, the obvious repair, which scores the reduced trace overlap, is provably capped below unity at the dominant eigenvector rather than the target (Sec. II). The ceiling is set by the objective, not by the optimizer.

We remove both by *purifying* the bath (Fig. 1). A thermofield ancilla $A \cong E$ keeps the enlarged $S+E+A$ state pure, so the pure-state fidelity applies exactly on the joint state, and the physical ρ_S is recovered afterward by tracing out E and A . This makes each projection a well-posed pure-state fit (cf. purified tensor networks [2]) and splits the difficulty into two questions. What reduced-state fidelity can

a fixed-depth purified ansatz attain, and is that ceiling set by expressivity alone? And how fast can each projection be driven to that ceiling, or can the optimizer itself limit the fidelity actually attained? Our contributions answer both. The purified formulation of Fig. 1 makes the attainable fidelity a property of the ansatz alone (Sec. II), and a Riemannian Gauss–Newton inner loop reaches it in a few iterations per projection (Sec. III).

II. PURIFIED JOINT-STATE PROJECTION

For a mixed target the reduced trace overlap obeys

$$\max_{\sigma_S \geq 0, \text{Tr } \sigma_S = 1} \text{Tr}[\rho_S^{\text{target}} \sigma_S] = \lambda_{\max}(\rho_S^{\text{target}}) < 1, \quad (1)$$

and the maximizer is the dominant eigenvector, so fitting the reduced state alone converges to the *wrong object*. A well-posed formulation must fit a pure object.

At inverse temperature β , the ancilla loads the Boltzmann amplitudes so that $E+A$ carries the thermofield double [3]

$$|\Omega_{EA}(\beta)\rangle = \sum_n \sqrt{p_n} |n\rangle_E |n\rangle_A, \quad p_n = \frac{e^{-\beta E_n}}{Z}, \quad (2)$$

whose bath marginal $\text{Tr}_A |\Omega_{EA}\rangle\langle\Omega_{EA}|$ is exactly the thermal state. The ansatz prepares a pure joint state $\rho_{SEA}(\theta)$ with A inert. Each p-VQD step propagates it to a target $|\Phi_t\rangle$ and refits the ansatz by the genuine pure-state fit

$$\min_{\Delta\theta} \frac{1}{2} \|\rho_\Phi - \rho_{SEA}(\theta_t + \Delta\theta)\|_F^2 = 1 - |\langle\Phi_t|\Psi(\theta_t + \Delta\theta)\rangle|^2. \quad (3)$$

The reduced state $\rho_S = \text{Tr}_{EA} \rho_{SEA}$ is read out only *after* the fit and is never optimized, precisely to avoid the cap (1). Performance is the trajectory average $\bar{F}$ of the reduced Uhlmann fidelity against the exact reference, so any deficit from unity comes from the ansatz or the optimizer, never from the objective.

III. RIEMANNIAN GAUSS–NEWTON INNER LOOP

With the objective fixed, the optimizer controls only the projection speed. Stacking the residual $R = \rho_\Phi - \rho_{SEA}(\theta)$ into real form with Jacobian J , linearization yields the damped normal equations

$$(J^\top J + \mu I) \Delta\theta = -J^\top r, \quad (4)$$

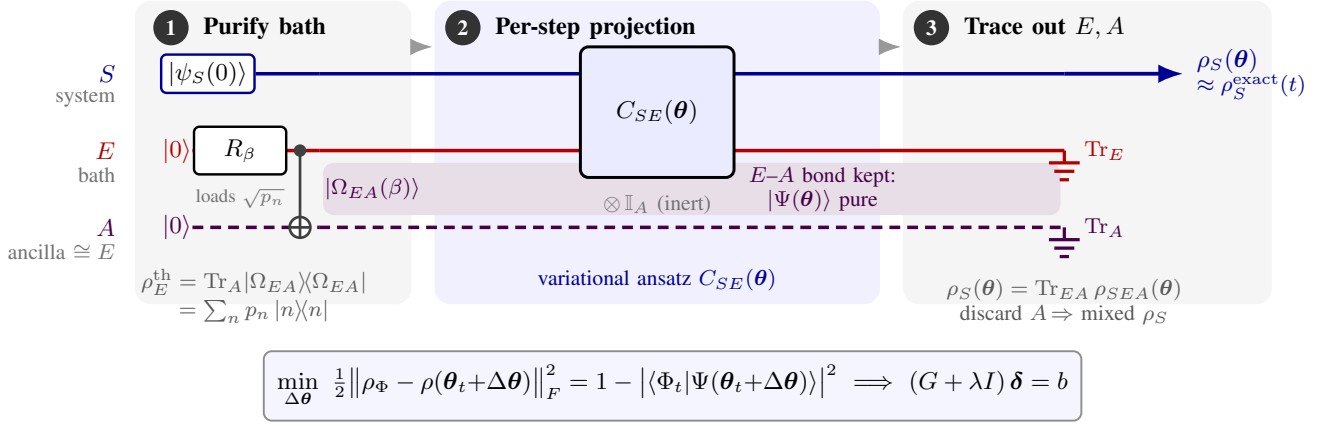


Fig. 1. Thermofield-purified p-VQD. An ancilla $A \cong E$ purifies the bath so the joint $S+E+A$ state stays pure, each step refits $C_{SE}(\theta)$, and E, A are traced out for the physical ρ_S .

solved by one Cholesky factorization. For pure joint states $J^\top J$ is the Fubini–Study metric [4], which makes the step *Riemannian* and rescales the ill-conditioned directions that stall gradient descent. A **Trust-Region** variant adapts the damping μ [5].

IV. RESULTS AND IMPACT

Two finite-temperature benchmarks, both exactly simulable after purification, separate the two questions. An independent McLachlan TDVP baseline [6], [7] flows the same purified ansatz continuously under the metric of Sec. III and is scored against the same reduced-state reference.

The *spin–boson* model [8] couples a qubit to one thermal bosonic mode,

$$H_{\text{SB}} = \frac{\Delta E}{2} \sigma_z + V_x \sigma_x + \omega a^\dagger a + \frac{g}{2} \sigma_z \otimes (a + a^\dagger), \quad (5)$$

Fock-truncated and purified by an isospectral ancilla. It is the controlled validation case. We compare three ansatz families of graded expressivity, namely HVA [9], ECD [10], and Polaron+HEA [11], [12]. The attainable fidelity is set by the ansatz alone. The Hamiltonian-matched HVA saturates the objective ($\bar{F} = 1.0000$) at every temperature, the polaron-dressed HEA nearly does ($\bar{F} \geq 0.99$), and the less expressive ECD is capped at 0.90–0.91. TDVP lands on the same per-family ceiling to within 3×10^{-3} , the signature of a well-posed objective. The optimizer then sets only the speed. Gauss–Newton and Trust-Region reach machine precision within 2–3 inner iterations, while the first-order loops need orders of magnitude more and plateau. At an equal iteration budget the second-order methods hold the ceiling and the first-order ones drift, with full-horizon $\bar{F}$ of 0.95 for GN against 0.69 for GD.

The *central-spin* model [13] couples a central qubit to N_b thermal bath qubits,

$$H_{\text{CS}} = \frac{\omega_0}{2} \sigma_z^S + \Delta \sigma_x^S + \sum_k \frac{b_k}{2} \sigma_z^S B_k + \sum_k \frac{g_k}{2} (\sigma_+^S \sigma_-^{B_k} + \text{h.c.}), \quad (6)$$

with each bath qubit purified by one ancilla, giving a genuinely decohering, non-Markovian stress test. The matched HVA again climbs to essentially unit fidelity with depth ($\bar{F} =$

0.9999 at $L = 16$) and coincides with TDVP wherever the ansatz is adequate, so expressivity again sets the ceiling. The generic HEA stays capped near 0.94 at every depth tried, and there the optimizers finally separate. The choice of optimizer now limits the *attained* fidelity rather than just the rate. The second-order steps attain the best tested fidelities and gradient descent lags well below.

The upshot is a drop-in open-system workflow. Purify the bath, choose an ansatz expressive enough for the target fidelity, and reach it cheaply with the Gauss–Newton and Trust-Region inner loop. Larger baths, hardware estimators, and shot-noise-aware damping remain open.

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